SIMATS ENGINEERING

ASSESSMENT TOOL 3

COURSE CODE: MLA03

COURSE NAME: Reinforcement learning for Environment and Agent

COURSE OUTCOME COVERED

CO4: Implement policy-based reinforcement learning methods to develop efficient solutions for complex environments. (BL2, BL3, BL6)

Assessment Tool Weightage: 30%

Dr G A. SENTHIL

Annexure A

Mock Interview

Total Marks: 50 Duration: 1hr

1. A machine learning engineer applies Reinforcement Learning to solve sequential decision-making problems. Explain how an agent learns optimal behavior through interaction with the environment. (10 Marks)

Answer:

- **Agent-Environment Closed Interaction Loop:** At each discrete time step t, the agent observes the environment state $S_t \in S$. Conditioning on its internal policy $\pi(a|s) = P(A_t = a \mid S_t = s)$, the agent samples and executes action $A_t \in A$. The environment transitions to a new state $S_{t+1} \sim P(s' \mid S_t, A_t)$ and generates a scalar feedback reward $R_{t+1} \in \mathbb{R}$. This state-action-reward cycle continues iteratively until terminal criteria are reached.
- **Credit Assignment & Cumulative Return Optimization:** The agent aims to maximize total accumulated discounted reward over time: $G_t = R_{t+1} + \gamma R_{t+2} + \gamma^2 R_{t+3} + \ldots = \sum_{k=0}^{\infty} \gamma^k R_{t+k+1}$, where $\gamma \in [0, 1)$ is the discount factor ensuring convergence and balancing short-term immediate payoffs vs. long-term strategic utility.
- **Recursive Value Function Modeling:** The agent models expected future returns using state-value $V\pi(s)$ and action-value $Q\pi(s, a)$ functions through Bellman Expectation Equations:
  • $V\pi(s) = E_\pi [G_t \mid S_t = s] = \sum_a \pi(a|s) \sum_{(s', r)} P(s', r \mid s, a) [r + \gamma V\pi(s')]$
  • $Q\pi(s, a) = E_\pi [G_t \mid S_t = s, A_t = a] = \sum_{(s', r)} P(s', r \mid s, a) [r + \gamma \sum_{(a')} \pi(a'|s') Q\pi(s', a')]$
- **Balancing Exploration and Exploitation:** To avoid sub-optimal local convergence, the agent balances exploiting known high-reward actions with exploring uncharted trajectories using strategies such as ε-greedy ($a = \arg\max Q(s,a)$ with probability $1-\varepsilon$, random action with probability ε), Boltzmann softmax exploration, Upper Confidence Bound (UCB), or entropy regularization bonuses.
- **Policy Optimization & Convergence to Bellman Optimality:** Using Temporal Difference (TD) learning or policy gradient ascent ($\nabla_\theta J(\theta) = E_\pi [\nabla_\theta \log \pi_\theta(a|s) Q^\pi(s, a)]$), the policy updates iteratively until satisfying the Bellman Optimality Equation: $V^*(s) = \max_a \sum_{(s', r)} P(s', r \mid s, a) [r + \gamma V^*(s')]$, arriving at the optimal decision policy π^*.

2. If you are asked to design a Reinforcement Learning solution for a real-time system such as ride-sharing or delivery optimization, explain how you would formulate the problem using a Markov Decision Process. (10 Marks)

Answer:

- **MDP Mathematical Formulation:** A real-time ride-sharing or delivery dispatch optimization problem is formulated as a formal Markov Decision Process tuple (S, A, P, R, γ).
- **State Space (S):** Encapsulates dynamic spatio-temporal variables across the system:
  • Driver/Vehicle Sub-State: Real-time GPS coordinates (xi, yi), operational status (idle, en route, occupied), battery/fuel levels, and current vehicle carrying capacity.
  • Demand & Order Queue Sub-State: Origin coordinates, destination drop-offs, elapsed customer wait times, surge pricing tiers, and delivery time-window constraints.
  • Environmental Context: Road traffic congestion heatmaps, weather condition indices, historical ride request densities, and time-of-day features.
- **Action Space (A):** Executed by the central platform or decentralized fleet at decision epoch Δt:
  • Dispatch Matching: Assigning available idle driver i to customer trip request j via bipartite matching algorithms.
  • Proactive Repositioning: Relocating unassigned vehicles from low-demand regions to high-probability future surge clusters.
  • Order Batching & Routing: Holding delivery requests for short consolidation windows to assign optimal multi-stop delivery routes.
- **Transition Dynamics (P):** Stochastic transition model P(St+1 | St, At) representing external real-world stochasticity such as traffic signal delays, Poisson customer order arrivals, dynamic speed variations, and order cancellation events.
- **Reward Function (R):** Multi-objective reward formulated to balance throughput, platform revenue, and service quality: Rt = $\alpha \cdot \sum$(Trip Fares & Delivery Revenue) - $\beta \cdot \sum$(Fuel Consumption + Deadhead Mileage) - $\lambda \cdot \sum$(Customer Wait Times + Order Delays) - $\mu \cdot$ (Cancellation Penalties).
- **Discount Factor (γ):** Configured near $\gamma = 0.95 - 0.99$ to ensure the policy values long-term fleet efficiency, network balancing, and future demand coverage over immediate short-distance trips.

3. Compare model-free and model-based Reinforcement Learning approaches. Explain their working principles and how they handle uncertainty. (10 Marks)

Answer:

- **Working Principles & Optimization:** • Model-Free RL: Optimizes policies $\pi(a|s)$ or action-value functions Q(s, a) directly through continuous trial-and-error environment interactions without building an explicit transition model (e.g., PPO, Soft Actor-Critic (SAC), DQN). Updates are driven strictly by sampled Bellman errors or policy gradients.
  • Model-Based RL: Explicitly learns an internal predictive world model consisting of transition dynamics $T\_\theta(s, a) \rightarrow s'$ and reward function $R\_\varphi(s, a)$. It performs planning and policy optimization over imagined trajectories via Model Predictive Control (MPC), Dyna architectures (MBPO), or Monte Carlo Tree Search (AlphaZero).
- **Sample Efficiency vs. Computational Overhead:** Model-free methods have low sample efficiency and require millions of environment steps to converge, but execute rapidly during

deployment. Model-based algorithms achieve superior sample efficiency by augmenting training data with synthetic rollouts, but require heavy computation during planning.

- **Handling Epistemic Uncertainty (Model / Parameter Ignorance):** • Model-Free: Uses intrinsic exploration bonuses such as Random Network Distillation (RND), count-based exploration, or entropy bonuses to encourage exploration of unfamiliar states.
  • Model-Based: Uses ensembles of probabilistic transition models (e.g., PETS, deep bootstrap ensembles) to quantify epistemic variance across networks, penalizing simulated paths where model variance is high to avoid model exploitation.
- **Handling Aleatoric Uncertainty (Inherent Environmental Stochasticity):** • Model-Free: Averages stochastic variations over large Monte Carlo trajectory rollouts or models the entire return distribution using Distributional RL (QR-DQN, IQN).
  • Model-Based: Generates Gaussian transition distributions $s' \sim N(\mu\_\theta(s,a), \Sigma\_\theta(s,a))$ and applies chance-constrained MPC to plan trajectories resilient to stochastic perturbations.

4. A recommendation system using Reinforcement Learning is underperforming. Explain how reward sparsity and hyperparameter tuning can be diagnosed and improved. (10 Marks)

Answer:

- **Diagnosing and Improving Reward Sparsity:** • Diagnosis: Millions of user impressions result in very few purchase/conversion events (<1%), causing value functions to collapse to zero, vanishing policy gradients, and poor feature learning in embedding layers.
  • Potential-Based Reward Shaping: Introducing intermediate dense feedback using potential functions: $R\_shaped(s, a, s') = R\_sparse + \gamma\Phi(s') - \Phi(s)$, where potential Φ assigns graded scalar rewards for intermediate engagements (e.g., item click: +0.1, dwell time > 30s: +0.3, add-to-cart: +0.6, checkout: +1.0). The potential formulation mathematically guarantees policy invariance.
  • Hindsight Experience Replay (HER) / Goal Relabeling: Sessions with unpurchased items are relabeled by setting the final browsed item as the virtual target goal, converting failed trials into informative gradient updates.
- **Diagnosing and Improving Hyperparameter Tuning:** • Learning Rate (α): Monitored via TD-loss oscillations. Addressed using AdamW optimizers with initial warm-up and Cosine Annealing decay to stabilize user/item embedding representations.
  • Discount Factor (γ): A small γ leads to myopic clickbait policies. Setting $\gamma = 0.98$ aligns optimization with long-term customer lifetime value (LTV) and multi-session retention.
  • Entropy Regularization: Prevents 'filter bubble collapse' and recommendation fatigue by tuning the policy entropy coefficient β, forcing broad catalog exploration.
  • Prioritized Experience Replay (PER): Rebalances training buffers by oversampling rare positive conversion transitions using TD-error priority weighting.

5. If you are implementing an RL-based game-playing AI, describe how you would design the learning pipeline and ensure training stability. (10 Marks)

Answer:

- **Learning Pipeline Architecture:** • Observation Preprocessing: Frame stacking (4 consecutive frames) to capture movement velocity and direction, grayscale downsampling to 84×84, pixel

normalization to [0, 1], and frame skipping (action repeat of 4).
• Vectorized Environment Simulation: Asynchronous parallel workers (e.g., Ray, SubprocVecEnv) to sample trajectories concurrently, decorrelating sequential transitions and maximizing GPU rollout throughput.
• Actor-Critic Deep Neural Architecture: Shared Convolutional/ResNet backbone feeding into dual heads: a policy head outputting action probabilities $\pi_\theta(a|s)$ and a value head estimating baseline state value $V_\varphi(s)$.

- **Ensuring Training Stability and Preventing Policy Collapse:** • PPO Clipped Surrogate Objective: Constrains step updates to prevent catastrophic policy collapses: $L_{CLIP}(\theta) = E_t [\min(r_t(\theta) \hat{A}_t, \text{clip}(r_t(\theta), 1-\varepsilon, 1+\varepsilon) \hat{A}_t)]$, where $r_t(\theta) = \pi_\theta(a_t|s_t) / \pi_{old}(a_t|s_t)$ and $\varepsilon = 0.2$.
  • Generalized Advantage Estimation (GAE): Balances bias and variance in policy gradients using exponential return weighting: $GAE(\gamma, \lambda) = \sum(l=0 \text{ to } \infty) (\gamma\lambda)^l \delta_{t+l}^V$.
  • Gradient Clipping & Orthogonal Initialization: Restricting global gradient norms ($||g|| \leq 0.5$) to prevent exploding gradients, paired with orthogonal weight initialization.
  • Running Normalization: Maintaining running mean and standard deviation for inputs and value targets to ensure stable gradient backpropagation.

Code of Conduct:
I K. Balamurali (192472010) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: K.Balamurali

RUBRICS FOR CALCULATION

Criteria	Weightage	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)
Technical Knowledge	25	Demonstrates strong and accurate subject knowledge with in-depth understanding	Shows good knowledge with minor gaps	Basic knowledge with noticeable errors	Lacks fundamental technical knowledge
Problem Solving **Analytical Thinking**	25 15	Solves problems logically and applies concepts effectively in real scenarios Analyzes questions critically and provides well-structured, logical answers	Applies concepts with moderate accuracy Provides reasonable analysis with some structure	Limited problem-solving ability; requires guidance Superficial or partially structured responses	Unable to solve or apply concepts No analytical thinking demonstrated
Communication	15	Communicates clearly, confidently, and professionally with proper terminology	Communicates well with minor hesitation	Communication lacks clarity or confidence	Unable to communicate effectively
Confidence **Response to Questions**	15 10	Displays high confidence, positive attitude, and professional behavior throughout Responds accurately, promptly, and elaborates with relevant examples	Shows good confidence with minor issues Responds correctly but with limited depth	Low confidence; inconsistent behavior Partial responses; needs prompting	Lacks confidence and professionalism Unable to respond appropriately